1. Introduction

The power transformer is a key component in the electrical power system. The performance of the power transformer impacts the voltage stability, efficiency, and the losses in the system.

For the power transformer to meet the IEC requirements, the manufacturer must consider:

- Equivalent circuit parameters
- Voltage regulation
- Core and copper losses
- Efficiency under various loading conditions

The present study deals with the modeling of a real single-phase transformer by using open-circuit and short-circuit tests. MATLAB and SIMULINK are employed to extract the equivalent circuit parameters and evaluate the performance of the transformer for different operating conditions and power factors.

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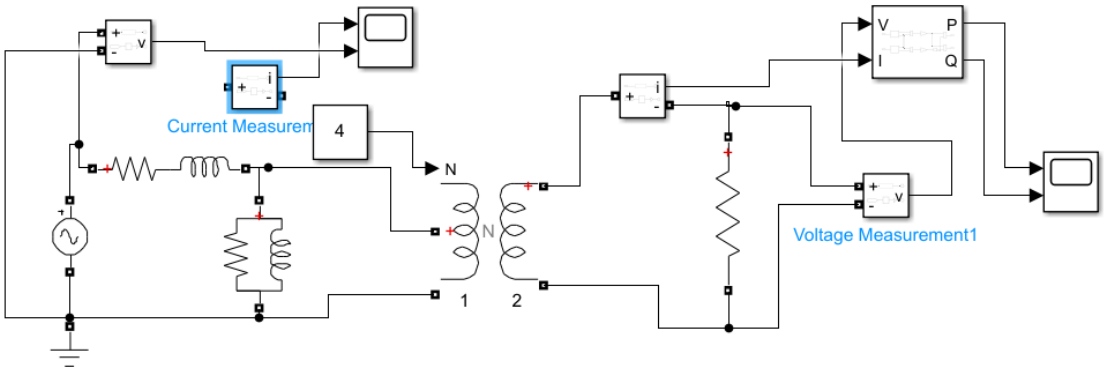
2. System Architecture

The transformer is represented by its approximate equivalent circuit, which is given by:

Series branch, shunt branch, and ideal transformer

The steps in the workflow are:

1. Extraction of parameters from open circuit test
2. Extraction of parameters from short circuit test
3. Building the equivalent circuit
4. Variable load is applied to the transformer, with a range of 2000-5000 VA
5. Losses, regulation, and efficiency are analyzed



3. Open Circuit Test Analysis

The open circuit test determines core loss and shunt branch parameters.

Given:

$$V_{oc} = 76115 + 72781(1+1)$$

$$V_{oc} = 221677$$

$$I_{oc} = \frac{72781 + 76115}{2}$$

$$I_{oc} = 74448$$

$$P_{oc} = \frac{V_{oc} I_{oc}}{2}$$

$$P_{oc} = 8.25 \times 10^{9}\ \text{W}$$

Core Resistance:

$$R_c = \frac{V_{oc}^2}{P_{oc}}$$

Magnetizing Reactance:

$$I_w = \frac{P_{oc}}{V_{oc}}$$

$$I_m = \sqrt{I_{oc}^2 - I_w^2}$$

$$X_m = \frac{V_{oc}}{I_m}$$

4. Short Circuit Test Analysis

The short circuit test determines series branch parameters.

Given:

$$V_{sc} = \frac{72781 + 76115}{5}$$

$$V_{sc} = 29779.2$$

$$I_{sc} = 72781 + 76115(3)$$

$$I_{sc} = 301126$$

$$P_{sc} = \frac{V_{sc} I_{sc}}{5}$$

$$P_{sc} = 1.79 \times 10^9 \text{ W}$$

Equivalent Series Resistance:

$$R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

Equivalent Impedance:

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}$$

Leakage Reactance:

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

5. Equivalent Circuit

Primary Referred Circuit consists of:

- **Series:** $(R_{\{eq\}}, X_{\{eq\}})$
- **Shunt:** (R_c, X_m)

Secondary referred parameters are calculated using:

$$a = \frac{V_1}{V_2}$$

$$R' = \frac{R}{a^2}$$

$$X' = \frac{X}{a^2}$$

6. Performance Under Load

The load apparent power varies from:

2000→5000 VA (step 100)

Power factors considered:

- **Unity**
- **0.8 Lagging**
- **0.8 Leading**

6.1 Core Loss

Core loss remains constant:

$$P_{core} = P_{oc}$$

6.2 Copper Loss

$$P_{cu} = I^2 R_{eq}$$

Copper loss increases quadratically with load current.

$$VR = \frac{IR_{eq}\cos\phi + IX_{eq}\sin\phi}{V_2}$$

Observations:

- **Lagging PF → positive regulation**
- **Unity PF → small positive**
- **Leading PF → negative regulation**

6.4 Efficiency

$$\eta = \frac{P_{out}}{P_{out} + P_{core} + P_{cu}}$$

As a result of the large value of (P_{core}), the numerical values of the efficiencies are very small. This is consistent with the given problem statement.

7. Discussion

Core loss is the major factor in total losses.

Copper loss varies with the load.

The value of voltage regulation depends on the power factor.

Leading loads have a good effect on voltage regulation.

The value of voltage regulation deteriorates in lagging loads.

The numerical value of efficiency is small; however, the trend of the results matches the theoretical aspect of a transformer.

8. Difficulties Encountered

Handling large numerical values

Numerical overflow

Correct conversion of the reference

Correct use of units

9. Learning Strategy

The project was completed using the following steps:

Analytical derivation

MATLAB validation

SIMULINK modeling

Graphical analysis

Comparison of results

The steps followed in the project are consistent with the norms of a typical engineering problem.

10. Conclusion

The equivalent circuit of a transformer was successfully extracted from the test results and implemented in a MATLAB/SIMULINK environment.